

Name: _____ Date: _____

Period: _____

Practice

Every solution gets **checked**. Every inequality gets **graphed**. Every quadratic gets one side set to **zero** before anything else happens.

1. **Solve, and justify one step.** For each equation, give the solution and name one property of equality you used along the way.

Equation	Solution	One property you used
(a) $4x - 9 = 3x + 2$	$x =$ _____	_____
(b) $6(x + 2) = 4x - 2$	$x =$ _____	_____
(c) $\frac{x}{3} - 5 = 1$	$x =$ _____	_____
(d) $3(2x - 1) + 4 = 5x + 9$	$x =$ _____	_____

Check your answer to (d) in the original equation. _____

2. **How many solutions?** You are given only the *last line* of each student's work.

(a) $7 = 7 \rightarrow$ _____ (b) $x = -2 \rightarrow$ _____

(c) $4 = 9 \rightarrow$ _____ (d) $0 = 0 \rightarrow$ _____

Which of the four came from an **identity**? _____ From a **contradiction**? _____

3. **Literal equations.** Solve each for the variable named.

(a) $V = \ell wh$ solve for w _____

(b) $2x + 5y = 20$ solve for y _____

(c) $cx - 4x = d$ solve for x _____

Item (c) is different from the other two. What did you have to do first, and why?

4. **Solve and graph.** Shade each solution set. Open $\circ$ for $<$ and $>$, closed $\bullet$ for $\leq$ and $\geq$.

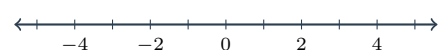
(a) $3x - 5 \leq 4$ x _____



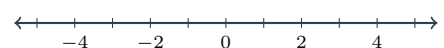
(b) $-2x + 1 > 9$ x _____



(c) $5(x + 1) \geq 3x + 7$ x _____



(d) $2(x - 1) < x + 3$ x _____



Exactly one of those four made you reverse the symbol. Which, and at which step?

5. **Write it, solve it, say it.** Answer each in a sentence with units — not as “ $x = \text{something}$.”

- (a) A gym charges a \$25 sign-up fee plus \$12 per month. Ana has paid \$121 in total.

Equation: _____ Answer: _____

- (b) A club orders T-shirts: a \$45 setup fee plus \$8 per shirt, and the budget is **at most** \$300.

Inequality: _____ Answer: _____

6. **Quadratics — pick your method.** Two are square-root problems, two factor, and two need the formula (leave those in simplest radical form).

(a) $x^2 = 64$, $x =$ _____ (b) $3x^2 - 27 = 0$, $x =$ _____

(c) $x^2 - 2x - 24 = 0$, $x =$ _____ (d) $x^2 + 7x = -10$, $x =$ _____

(e) $x^2 - 6x + 2 = 0$, $x =$ _____ (f) $2x^2 + 4x - 1 = 0$, $x =$ _____

The two that need the formula — work them in full.

(e) $x^2 - 6x + 2 = 0$

(f) $2x^2 + 4x - 1 = 0$

Which item required a Step 0 the others did not, and what was it?

Verify one solution of (c) by substituting it back. _____

7. **Count before you solve.** Fill the table in *without* solving anything.

Equation	$b^2 - 4ac$	Positive / zero / negative?	# real solutions
(a) $x^2 + 5x + 6 = 0$	_____	_____	_____
(b) $x^2 + 4x + 4 = 0$	_____	_____	_____
(c) $x^2 + x + 5 = 0$	_____	_____	_____
(d) $3x^2 - 2x - 1 = 0$	_____	_____	_____

One of those four has a graph that touches the x -axis without crossing it. Which? _____

8. **Explain.** A classmate solves $(x + 1)(x - 4) = 6$ by writing “ $x + 1 = 6$ or $x - 4 = 6$.” Explain what is wrong, why the zero product property needs the zero, and then solve the equation correctly. _____

Extension optional

The solution-count audit. In each problem below, a letter sits *inside* the equation. Your job is to say which values of that letter produce each possible number of solutions — so the count becomes something you control rather than something you discover.

Number of solutions	For which c does $5x + c = 5x + 8$?	For which b does $x^2 + bx + 9 = 0$ have that many <i>real</i> solutions?
two	—	_____
exactly one	_____	_____
none	_____	_____
infinitely many	_____	—

- (1) One box in the left column is impossible to fill. Which, and why can that never happen for $5x + c = 5x + 8$? _____

- (2) A linear equation *can* have infinitely many solutions. A quadratic equation with $a \neq 0$ never can. Explain why not — what is it about the x^2 term that makes an identity impossible? _____

- (3) In the right column, the answers to “two” and “none” are ranges of b , not single values. Say what that means about the graph of $y = x^2 + bx + 9$ as b increases from 0 through 6 and past it. _____

Coming next — Lesson 1.3: Functions. The third and last Algebra 1 domain area, and today's equations become its machinery. A function's **zeros** are defined as the solutions of $f(x) = 0$ — so every technique in problem 6 is a way of finding where a graph crosses the x -axis, and the discriminant in problem 7 is a way of counting those crossings before you draw anything. The inequalities in problem 4 are how a function's **domain** gets written down when some inputs are not allowed. And the literal equations in problem 3 are the reason $y = mx + b$ has three different “forms” in Unit 2: it is one equation, solved for different things.